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DAY 5

DEEP LEARNING & NEURAL NETWORKS

The Engine Behind ChatGPT, Self-Driving Cars & Image Generation

A Course Guide by AYE Tech Hub

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1. Introduction

In Day 4 you learned that **Machine Learning** lets computers find patterns in data. Today you'll meet the technology that took those ideas and exploded them into the most transformative breakthrough in computer science of the last 50 years: **Deep Learning (DL)**. Deep Learning is what makes ChatGPT speak fluently, Tesla cars steer, Naver Papago translate, and your phone unlock by recognising your face.

The core idea sounds simple — stack many layers of artificial 'neurons' that pass signals to each other, and let the network adjust its own connections until it gets the right answer. But making this work at scale required three things to come together around 2012: **massive datasets**, **powerful GPUs**, and **better algorithms**. Once they did, deep learning began beating humans at task after task — vision, speech, translation, games, and now language itself.

Learning Objectives

By the end of this lesson you will be able to:

- Explain what an artificial neuron is and how it relates to a biological neuron.
- Describe the layered architecture of a neural network (input, hidden, output).
- Explain in your own words how 'backpropagation' lets a network learn from mistakes.
- Identify the three main neural-network families (CNN, RNN, Transformer) and what each is for.
- Recognise the major milestones that brought us from AlexNet (2012) to ChatGPT (2022).

2. From Biology to Computation

Deep learning began as an attempt to mimic the brain. Your brain contains about **86 billion neurons**, each connected to thousands of others through synapses. A neuron fires when enough of its inputs are active — a tiny electrical decision repeated billions of times per second. From this simple mechanism, all human intelligence emerges.

In 1943, neuroscientist Warren McCulloch and logician Walter Pitts proposed that a neuron could be modelled mathematically. An **artificial neuron** takes several numerical inputs ($x_1, x_2, x_3 \dots$), multiplies each by a **weight** ($w_1, w_2, w_3 \dots$), sums the results, adds a **bias**, and passes the total through an **activation function** that decides if the neuron 'fires'. The output goes to the next layer.

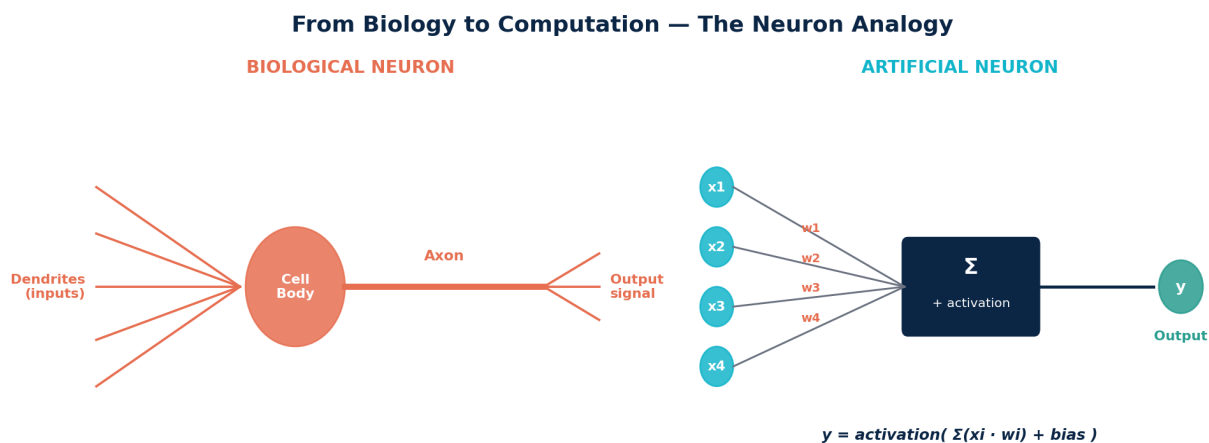


Figure 2.1 — A biological neuron (left) and its artificial counterpart (right). Both gather many input signals, combine them, and decide whether to send an output.

Key Insight

A single artificial neuron is mathematically very simple — just multiplication and addition. The power of deep learning comes from **stacking millions of these neurons together** and letting them adjust their weights through training. The model itself is not 'intelligent' — but the patterns it discovers in massive data can be astonishing.

3. Building a Neural Network

A single neuron can only solve very simple problems. The breakthrough comes when you connect many neurons into **layers** and stack the layers in sequence. The result is a **neural network** — the basic building block of all deep learning systems.

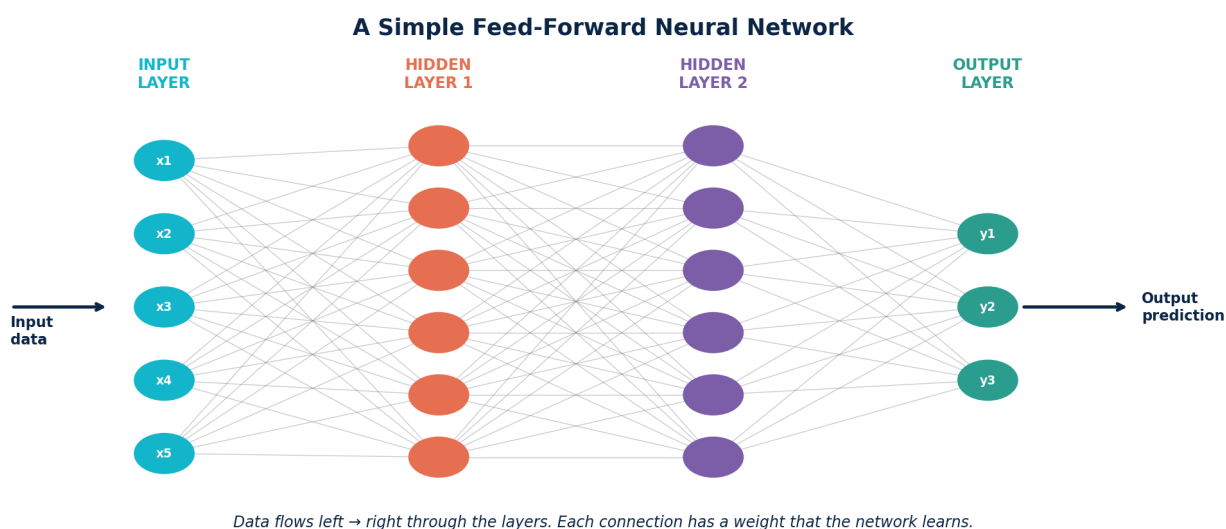


Figure 3.1 — A feed-forward neural network. Input data flows left to right through the hidden layers, with every neuron in each layer connected to every neuron in the next.

The Three Layer Types

Layer	Job	Example
Input layer	Receives the raw data	Pixels of an image, words of a sentence, sensor readings
Hidden layer(s)	Find patterns and combine features	Detects edges, then shapes, then objects (in images)
Output layer	Produces the final prediction	Class label, probability, predicted number

Weights, Bias & the 'Model'

Every connection between two neurons has a **weight** — a number that says how much that input matters. Every neuron also has a **bias** — a number that shifts its threshold. The 'model' is simply the collection of **all weights and biases** in the network. A small network may have a few thousand weights; **GPT-4 has over a trillion**. Training is the process of nudging every one of those numbers in the right direction.

4. Why 'Deep'? Shallow vs Deep Networks

A network with just one hidden layer is called **shallow**. A network with many hidden layers (often dozens or hundreds) is called **deep** — and that is where the name **Deep Learning** comes from. The difference is not just size; it is a difference in **what each layer can learn**.

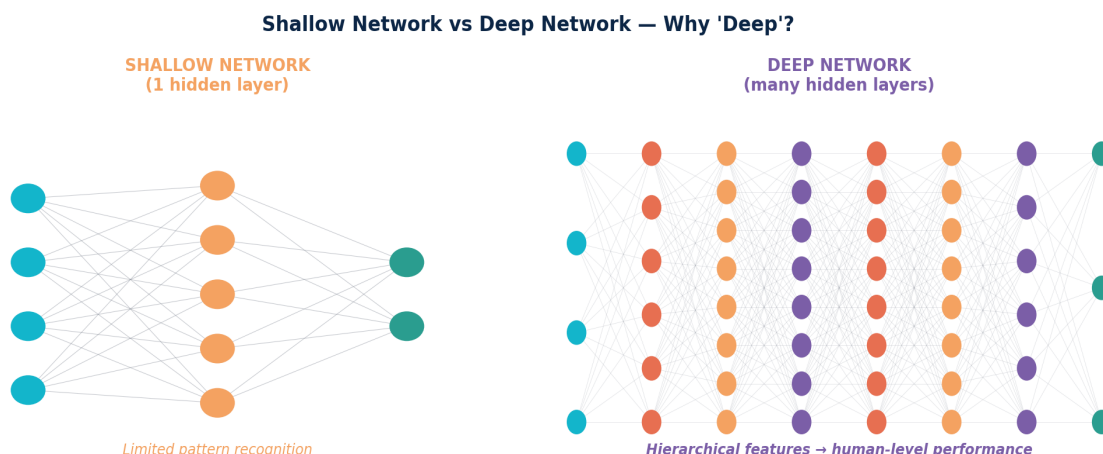


Figure 4.1 — A shallow network (left) has one hidden layer. A deep network (right) has many. Each additional layer can recognise more abstract patterns.

In a deep network, each layer learns features built from the output of the layer before it. In an image-recognition network, layer 1 might learn edges and colours, layer 2 combines those into shapes, layer 3 into object parts, and layer 4 into whole objects. This **hierarchy of features** is what made deep learning so much more powerful than any previous approach — and it is also why deep networks need huge datasets and lots of computing power to train.

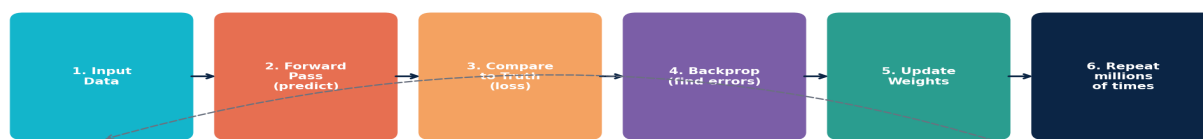
Real Numbers — Why GPUs Matter

Training a modern deep network involves trillions of multiplications. A high-end NVIDIA GPU performs ~80 teraflops (80 trillion operations per second). Training GPT-4 reportedly used **25,000 GPUs running for 90+ days** — at an estimated cost of \$100+ million. This is why only a handful of companies in the world can train the largest models. But **using** a trained model (called 'inference') is much cheaper — that is what makes ChatGPT free or low-cost for end users.

5. How a Neural Network Learns

If a network is just a bunch of weights, how do they get set to the right values? Through **training** — a loop that repeats millions of times. The network starts with random weights, makes terrible predictions, measures how wrong it was, and adjusts the weights slightly to reduce the error. Repeat enough times, and the network becomes remarkably accurate.

How a Neural Network Learns — The Training Loop



Each loop nudges weights toward better predictions — this is 'learning'

Figure 5.1 — The six steps of neural-network training, repeated millions of times until the network's predictions match the true answers closely enough.

Key Terms You Need to Know

Term	Meaning
Forward pass	Data flows through the network, producing a prediction
Loss function	A formula that measures how wrong the prediction was
Backpropagation	Algorithm that traces the error backward through the network, calculating how much each weight contributed
Gradient descent	Method for adjusting weights — take a small step in the direction that reduces loss
Epoch	One full pass through the entire training dataset
Batch size	How many examples are processed before updating the weights
Learning rate	How big a step to take when updating weights — too big = unstable, too small = slow

Engineer's Analogy

Think of training like adjusting a complex 3D control panel with millions of knobs. After every prediction, the system measures how wrong it was, then turns each knob a tiny amount in the direction that *should* reduce the error. Backpropagation is the algorithm that figures out, for each knob, which direction is correct. Do this billions of times and the knobs settle into a configuration that predicts well — even on data the network has never seen.

6. Activation Functions

An activation function decides what each neuron outputs given its weighted sum of inputs. Without activations, no matter how many layers you stack, the whole network would collapse into a simple linear formula. Activations add the **non-linearity** that allows the network to learn complex patterns.

Common Activation Functions — The Neuron's 'Decision'

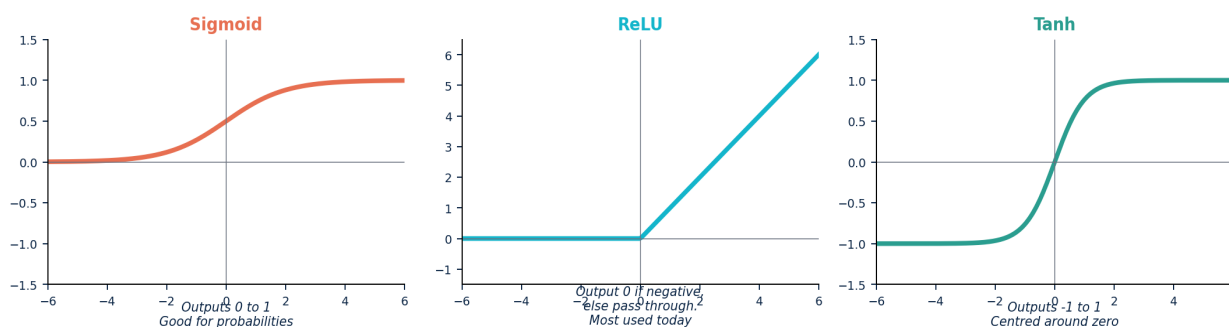


Figure 6.1 — The three most common activation functions. ReLU (centre) is the default choice in modern deep networks because it is simple, fast, and trains well.

7. Three Families of Deep Networks

Not all deep networks are built the same way. Three major architectures handle fundamentally different kinds of data — images, sequences, and language. Knowing which to use for which problem is one of the most important skills in modern AI engineering.

Three Major Families of Deep Networks

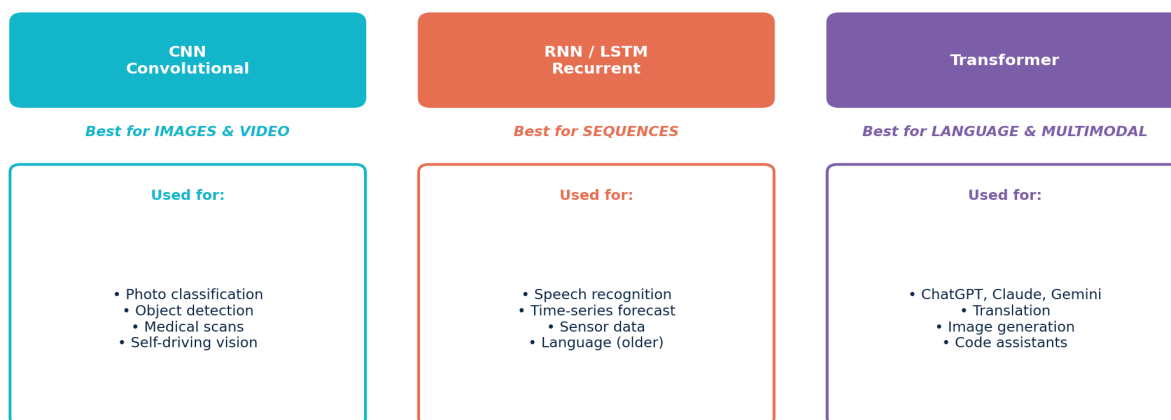
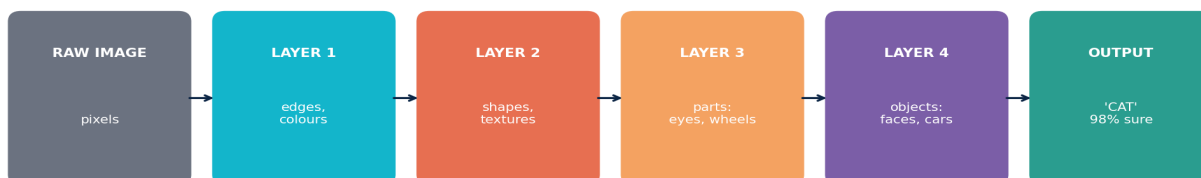


Figure 7.1 — The three families that dominate deep learning today: CNNs for vision, RNNs/LSTMs for sequences, and Transformers for language and multimodal AI.

8. CNNs — How a Network Sees an Image

Convolutional Neural Networks (CNNs) are designed for grid-like data — most famously images. A CNN does not look at every pixel as a separate input. Instead, small filters scan across the image looking for specific patterns: edges, corners, textures. Layer by layer, the network builds increasingly abstract concepts from these basic building blocks.

How a CNN 'Sees' an Image — Feature Hierarchy



Each layer learns more abstract features — from pixels to concepts

Figure 8.1 — How a CNN processes an image. Early layers detect simple features (edges, colours); deeper layers detect complex concepts (faces, vehicles, animals).

Real Applications of CNNs

- **Photo recognition** — Google Photos, Apple Photos search by 'beach', 'dog'.
- **Medical imaging** — detect tumours in X-rays, retinal disease in eye scans.
- **Quality control** — factory cameras inspect every part for defects in real time.
- **Self-driving cars** — recognise lanes, pedestrians, traffic signs, other vehicles.
- **Face ID** — your phone unlocking by recognising your face under any lighting.

9. The Deep Learning Revolution

Deep learning has been around in theory since the 1980s — but it only became practical in 2012, when a team led by Geoffrey Hinton used a deep CNN called **AlexNet** to win the ImageNet competition by a huge margin. The win shocked the AI field and triggered an explosion of research and investment that continues to

this day.

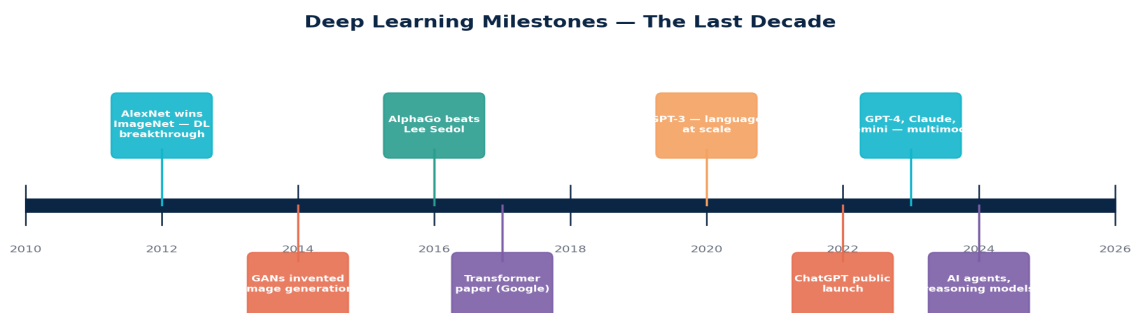


Figure 9.1 — Major deep-learning milestones from 2012 to 2024. Each breakthrough opened new applications and triggered the next wave of progress.

10. Summary, Self-Check Quiz & Homework

Lesson Recap

- **Deep Learning** = neural networks with many hidden layers, inspired by the brain.
- A **neuron** = weighted sum of inputs + bias + activation function.
- A **network** = many neurons organised into input, hidden, and output layers.
- **Training** uses forward pass, loss measurement, backpropagation, and gradient descent — repeated millions of times.
- **ReLU** is today's default activation function; sigmoid and tanh are older alternatives.
- Three main families: **CNN** (images), **RNN/LSTM** (sequences), **Transformer** (language and multimodal).
- The 2012 **AlexNet** win triggered the deep-learning revolution that gave us ChatGPT, Tesla autopilot, and image generation.

Self-Check Quiz (answers at the bottom)

1. What are the three layer types in a neural network?
2. What does 'deep' mean in 'deep learning'?
3. Which family of network is best for image classification?
4. Backpropagation is used to: (a) make predictions, (b) update weights based on error, (c) increase the number of layers, (d) store data.
5. Which event in 2012 is often called the 'birth of modern deep learning'?

Answers: 1) Input, hidden, output. 2) The network has many hidden layers stacked in sequence. 3) Convolutional Neural Network (CNN). 4) (b) update weights based on error. 5) AlexNet winning the ImageNet competition.

Homework Assignment

Pick one deep-learning product you use daily (face unlock, Google Translate, Spotify auto-mix, Naver Papago, Tesla autopilot demos, or ChatGPT) and complete the following:

1. Identify which **family of network** (CNN, RNN/LSTM, or Transformer) it most likely uses, and explain why.
2. List three **input features** the network receives.
3. What is the **output** (label, sentence, action)?
4. Estimate how the network might **fail** if your input is unusual. Give one realistic example.
5. Sketch a rough block diagram of the system (input → network → output).

Submit to awetgknway@gmail.com with subject *AI Day 5 Homework — Your Name*.



Coming Up Tomorrow — Day 6

Generative AI — the technology that creates images, music, video, and human-like text from a few words of prompt. You'll learn how DALL·E, Midjourney, Sora, and ChatGPT actually *generate* new content instead of just classifying existing content.

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